```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/Midterm_Part_2$ ./sobel_pthread NGGYU.jpg 8
Loading color image: NGGYU.jpg...
Running Sobel operator sequentially...
Time taken sequentially: 0.014163 seconds
Running Sobel operator with 8 pthread(s)...
Time taken with pthreads (8 threads): 0.003010 seconds
Parallel output matches sequential: YES
Saved edge visual to 'output_sobel_pthread.bmp'
```

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/Midterm_Part_2$ ./sobel_openmp NGGYU.jpg 8
Loading color image: NGGYU.jpg...
Running Sobel operator sequentially...
Time taken sequentially: 0.012442 seconds
Running Sobel operator with 8 OpenMP thread(s)...
Time taken with OpenMP (8 threads): 0.003014 seconds
Parallel output matches sequential: YES
Saved edge visual to 'output_sobel_openmp.bmp'
```

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/Midterm_Part_2$ ./sobel_pthread NGGYU2.jpg 8
Loading color image: NGGYU2.jpg...
Running Sobel operator sequentially...
Time taken sequentially: 0.312966 seconds
Running Sobel operator with 8 pthread(s)...
Time taken with pthreads (8 threads): 0.048839 seconds
Parallel output matches sequential: YES
Saved edge visual to 'output_sobel_pthread.bmp'
```

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/Midterm_Part_2$ ./sobel_openmp NGGYU2.jpg 8
Loading color image: NGGYU2.jpg...
Running Sobel operator sequentially...
Time taken sequentially: 0.304797 seconds
Running Sobel operator with 8 OpenMP thread(s)...
Time taken with OpenMP (8 threads): 0.048607 seconds
Parallel output matches sequential: YES
Saved edge visual to 'output_sobel_openmp.bmp'
```